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Phase 1 Verifying and cleaning

Step 1 - Verifying the parts

It is generally a good idea throughout the building of the bike to make sure the parts don't show any errors and/or fit properly during building.

Parts needed to start off;

- 1x Main frame (0074)
- 1x lower-wishbone-rear R (0147)
- 1x lower-wishbone-rear L (0456)
- 1x lower-wishbone front L (0070)
- 1x lower-wishbone front R (0454)
- 4x upper-wishbone F/R L/R (0146)
- 2x kingpin rear L/R (0071)
- 1x kingpin front left (0072), 1x kingpin front right (0073)
- File and clean where steering bearings will go. Do this with care; hence slightly inwards angle on inner edge to leave no visible trace)

Phase 2; Pre-assembly and preparing parts

Step 7 - Preparing the varied rod ends for wheels suspension

- select & prepare the different joints
- 2x rod end M8 left thread (0018) - 6x rod end M8 right thread (0019) - 4x rod end M10 (0017)
- grease and brush
- workbench vise
- bellow for rod end M10 (0021)
- bellow for rod end M8 (0022)
- 4x M10 regular nut
- 8x M8 regular nuts
- small tiwraps
- insert a rod end into side-slit (not bottom!) of appropriate size bellow
- grease the ball joint generously with special attention to the moving parts
- use vise to clamp down the rod end
- use nose-plier to gently pull the bellow over the rod end
- repeat for all rod ends (remember the left-threaded rod ends)
- screw on appropriate nuts (not nylon-stop nuts) on ball-joints without forcing the bellows
- use small tiwraps (100mm) to fix rubber covering in place. Check carefully if exit slits align. Be precise.

keep finished product clean with paper

Step 8 - Steering rod (0055) pre-assembly

- select and use the 2x rod end M8 left thread and 2x rod end M8 right thread plus 2 steering rods screw joints in their respective places but don't tighten yet

Step 9 - Attaching remaining rod ends in correct positions

- Get 1x rod end M8 right thread
- Get the rear kingpin

Materials;

- M8x40 socket head cap
- 2x distance bushes for RERS2 M8 (0025)
- 1x washer M8 (8,4x16)
- 1x nylon stop nut (M8)

- assemble the rod end and distance bushes and push in place
(*tip; use screwdriver and/or nose-needle plier to align and position the ball-joint and its washer into correct position for the bolt)
- put in the bolt, finish with the washer and nylon stop. Tighten accordingly.
- repeat and the other rear-kingpin but mirror the bolt for later alignment on the bike.

Step 10 - Preparing and attaching rod end M10 on rear kingpin

- Get 2x rod end M10
- Materials (for one rear kingpin);
- 2x M10x45 socket head cap bolt
 - 4x distance bushes for RERS2 M10 (0024) optional at this stage;
 - 4x washers M10 [10,5x20]
 - 2x nylon stop nut M10
 - Insert and align a bolt and 2 rubber-spacer-washers into ball-joint cover
 - screw bolt in place on rear-kingpin (20Mn torque)
 - repeat
 - repeat on other rear-kingpin

Step 11 - Attaching rod end M8 to front upper wishbones & preparation for front kingpin attachment

Upper wishbone pre-attachment prep for 1 (!) Materials;

- 1x nylon stop nut M8
- 2x M8 washer
- fix slide bearings black PA (0179) into ALL upper wishbones in preparation for later attachment to main frame
- Fix the rod end to the upper wishbones using washers and nut. Leave loose for future handling

Preparing for front kingpin attachment (x2) Materials;

- M8x40 bolt
- 1x washer M8
- 2x distance bushes
- 1x nylon stop nut
- insert in correct order and use nylon stop nut to keep in place. Leave loose for further handling

Step 12 - inserting steer bearings

- Get the correct bearing steering rod (0141) (4x)
- If step 1 is correctly done ((file and clean steering bearing holes), insert the bearing straight.
- use a rubber mallet or one smaller and big one to tap into place if tight fit
- if the bearing falls out keep aside and put in place at later handling
-

Step 13; build seat slider

Materials;

- 4x M8x35
- 8x M8x16 bolt
- 5x M8x16 countersunk torque screws
- 2x washers M8x24
- 2x (0033) seat slide profile
- 4x seat slide bearing (0032)
- 1x (0416) Slide stopper Right
- 1x (0415) Slide stopper Left
- 1x (0060) seat adjust handle bar
- 1x (0061) seat adjust handle grip
- 1x Connector block for Seat adjust handle bar (0076)
- 2x Connector blocks for Seat stop (0075)
- 2x Seat connector block (0086)
- 1x Seat rear support bracket (0048)
- 2x 0407 Seat Connection Aluminium L-profile

- 2x Extension Spring for Handle (0010)
- 2x Stop Ring for Extension Spring (0417)
- 2x slide bearing for seat connector block stop
- (2x rod for slide stopper [0059])

See pictures for assembly. Use weak loctite on every bolt.

- square the slide stoppers to 90°
- use vice to compress plastic slide bearing for seat connector

Step 14.1 Fixing handle bar to slide

Step 14.2 Extension Spring for handle installing (2x)

Materials;

- 2x M4x20 countersunk
 - 4x nylon stop nut M4
 - 2x extension ring
 - 2x key ring for closing (not mounted)
- M4 countersunk -> nylon stop nut >- spring <- nylon stop nut
 - Pull through pre-cut holes top and attach ring to spring-ends.

Step 15 - Lifting the bike on the struts

- Clean your workplace to keep the bike parts pristine
- You'll be fastening the bike with 2 M6-bolts diagonally relative to where the chaintube will be. So, concrete, no bolt where the chaintube will be installed (front fastening) and the second one diagonally from there.

2x Materials;

- 1xM6x40 bolt
- 1x washer 6 (6,4x12)
- 1x thick washer (6,4x17 KVP)
- 1x nylon stop nut M6
- fasten the bike to the back strut first then the front

Phase 3: Assembling the front kingpins and steering

Step 1 - installing steering lower end & preparings

Materials;

- 2x steering head (0044)
- 2x steering head top cap (0064)
- 2x steering head bottom cap (0181)
- 2x steering arm (0065)
- 1x steering arm connector bar (0066)
- 2x bearings for steering arm (0028)
- 2x retaining rings for steering arm (0140)
- 2x M6x12 countersunk screw
- install small bearings by gently pushing them into the 2 open spaces of the steering arm connector. Use the small retaining rings to lock in place with the help of appropriate pliers.
- push through the front steering holes the steering head from top. Don't forget to put the top cap on.
- if needed, insert remaining bottom bearing into appropriate opening if the fit wasn't tight enough.
- put on one steering arm (work it a bit beforehand if need be or slightly grease it)
- put on bottom cap and screw loosely in place (M6x12)
-

Step 2; Install steering arm connector

- fasten steering rods with M8x50 bolt, distance bushes M8 (DB), 2 washers (M8x16) and 1 nylon stop nut M8
- bolt -> DB -> rod end -> DB -> steering arm -> 2x washers
 (double) arm connecting bar -> nut - tighten accordingly

Step 3; seat slide profile attachment & slide installment Materials;

- 12x M8x16 bolts
- file down rail on the inside to accommodate the welt.
- Test if slide goes smoothly and then loctite in place. Use strong loctite (243).

Step 4; Install and assemble spider-joint on front kingpins (L/R), lower front wishbones and front upper wishbones.

Step 4.1 Attach lower front wishbones X2

Materials;

- 2x M8x50 bolts
- 2x washers M8
- 2x nylon stop nut M8
- install lower wishbone into prepared mounts, push the bolt through the silent block and tighten accordingly -

Step 4.2 Attach front kingpins to spider-joints X1

Materials;

- 2x M10x45
- 2x M10 washer
- 2x nylon stop nut M10
- move kingpin over spider-joint rod end and turn to align the kingpin and the rod end so the M10 nut can pass.
- fasten in place with bolt, washer and nut and tighten accordingly

Step 4.3 Attach upper wishbones to main frame

Materials;

- 4x M6x40
- 4x M6 washers
- 4x nylon stop nut M6
- mount upper wishbone into place and insert M6 bolts (remember to point the inward). Fasten accordingly.

Step 4.4 Attaching upper wishbones to front kingpin

Materials;

- (2x M8x40 bolt)
- 4x distance bushes
- 2x washer M8
- 2x nylon stop nut M8
- join together and tighten accordingly

Step 4.5 Attaching steering rod to front kingpins (top)

Materials;

- 2x M8x45 bolt
- 4x distance bushes
- 2x washer M8
- 2x nylon stop nut M8
- join steering rod to kingpins and tighten accordingly

Step 4.6 Attach brake adapters to front kingpins

Materials;

- 3x brake adapters
- 6x brake adapters black bolts pre-washers
- Install 2 front left kingpin (1; disc brake, 1; parking brake)
- Install 1 front right kingpin (1; disc brake)

Use weak loctite (222) to fasten bolt in place

Phase 4: Assembling & mounting freewheel and Bafang

Step 6; attach rear upper wishbone to main frame

Materials;

- 4x M6x40

- 4x M6 washers
- 4x nylon stop nut M6
- See front upper wishbone attachment & repeat

Step 7; Attach lower rear wishbones X2

Materials;

- 2x M8x50 bolts
- 2x washers M8
- 2x nylon stop nut M8

Step 8; Attach rear kingpins to lower rear wishbones Materials;

- see pre-assembly if not present
- attach and tighten accordingly but leave loose for cardan axle

Step 11; Shock Absorber Installation front and back (seperate it)

Materials;

- 4x shock absorbers (0142)
- 8x M8x40 bolt
- 8x M8 washers
- 8x M8 nylon stop nut
- install shock absorber at appropriate top holder main frame (use small allen key to straighten out slit)
- fasten with bolt, washer and nut
- fasten lower in middle hole
- repeat for all shock absorbers

Step 12; Installing rear axle & fastening rear kingpin to rear upper wishbones

Materials;

- 2x Rear wheel axle (0310)
- 2x Cross Cardan Shaft (0192)
- 4x Parallel Key 6X6X20 (0091)
- axle grease (C0012) (standard grease; C0013)
- 4x M8 washers
- 2x nylon stop nuts
- align the freewheel sprocket with the Rohloff sprocket, fasten the freewheel in place with the bearing small screws
- insert parallel keys into freewheel axle and grease
- tap the cardan shaft into place. Keep in mind and aligned. Keep the spring aside for practical reasons.
- insert rear wheel axles into rear kingpin bearings
- insert parallel keys into rear wheel axle and grease
- tap the other half of the cardan shaft into place
- reconnect (don't forget the spring!) the two parts while inserting the top rod end into the upper wishbone. Fasten it and tighten accordingly with the nylon stop nut and washers

Step 14; Attach brake adapters to rear kingpins

Materials;

- 2x IS brake adapter (0461)
- 4x brake adapters black bolts pre-washers
- Install them on the rear kingpins, facing back Use weak loctite (222) to fasten bolt in place

Phase 5 Rohloff ; Steering ; Brakes; wheel alignment

Step 1; install steer handles

Materials;

- 8x M6x20

- 2x steering handle bars (0462)
- use tube to align both steering heads
- align middle hole of steering arm connector with bike frame
- use M6x20 bolt to fix the heads in place, use weak loctite and gently equally tighten them
- insert handle bars and tighten gently and equally (so it doesn't crack), use the M6x20

Step 3; installing mudguard holders

Materials;

- 12x mudguard holder (0045)
- 24x M5x20
- 16x M5 nylon stop nuts
- 16x M5 washers
- screw top holders in place using weak loctite but no washer
- attach the other holders using nylon stop nut and washers, bolt pointing inwards
- for shipping purposes attach with outer bolt only so it can retract.

Step 4; Preparing disc brakes

Materials;

- 4x brake disc (0470)
- 4x Samagaga wheelhub (0077)
- 24x brake disc bolts M5x10mm (included with disc brakes) (0187)
- screw the disc brake on to the wheel holders 2x2 mirrored for left and right. Respect the drive direction.
- use weak loctite
- torque to 6Nm
- don't touch the disc brake outer edge with bare hands or grease

Step 5; Installing disc brakes on main frame

Materials;

- 4x stop screw hex M5x8
- 2x front wheel axles (0057)
- grease
- 2x parallel keys
- 2x front kingpin back cover (xxxx)
- place locking block in appropriate slit (back wheels)
- grease the axle
- slide over disc brake tightly (so no horizontal movement over the axle) and tighten with the hex stop screw
- insert front axle, grease and slide over disc brake, keep tight like rear and fix in place with stop screw over the cutout on the axle.
- install back cover on front kingpin.
- to get the disc brake close and tight use a wheel and tighten closely, then fix the stop screw.

Step 6; Installing & adjusting brake systems

Materials;

- 2x front brakes (0404)
- 2x back brakes (0376)
- 7x p-clips
- 1x 2cm screw 4,2mm
- 6x 1cm 4,2mm?
- align and fix the disc brakes on the discs. They should align logically.
- pull cable through middle of bike frame while attaching it using the pre-drilled smaller holes (3,5mm) and buntbandsset.
- repeat for front
- for adjusting see Tektro specifications (drain little oil if necessary)

Step 7; making the wheels (pre-assembly ->)

Materials;

- 4x inner tubes 20" 40/62-400 (0176)
- 4x Marathon Plus 20" tire (0177)
- 4x Alligt wheel (0397)
- 4x air valve rings (xxxx)
- 4x air valve caps (xxxx)

- install tube, tires on the wheel hub to standard specifications (not more than 5bar of air pressure)

Step 8; aligning wheels

Materials;

- use the aligning tools for this job
- 1 rod to keep steering heads in place
- 4x wedge lock washers HLS 10
- 4x M10x30
- 4x wheel caps
- align using the custom made aligning tools
- use the bolts, washers and caps to fasten the wheel loosely. Tighten them at the end.

Phase 6 Front boom, main chain, batteries & test drive

Step 10; Making & installing the seat (0,5h)

Materials;

- 1x M6x55 socket head cap bolt
- 1x M6 washer
- 1x nylon stop M6 nut
- 1x seat holder back bar
- 1x seat support bar front
- 1x seat
- 1x seat cushion
- double sided tape
- 2x countersunk screw M6x12
- 4x M6x12 bolt
- 2x seat slide-connectors
- fasten seat holder back bar with the M6x55 bolt, washer and nylon stop nut
- fasten seat support bar with the sunken screws and weak loctite
- mount the seat on the slider and fasten the seat slide connectors with the M6x12 nuts and loctite
- fasten de back bar
- tape the cushion to the seat with double sided tape (Tesla)

Step 13; moulding the mudguard using heatgun (pre-assembly->) (0,5h)

Materials;

- 4x mudguards
- mould them (remember placement on bike) with heatgun and round mall. Do this carefully as to not to burn the plastic.
- mount them or skip for shipping

Step 15; installing mirrors & preparing handles and installing (1,25h)

Materials;

- 2x mirrors
- 2x rubber handles
- 1x Rohloff adapter tube
- 1x left hand adapter tube
- put the rubber handles in warm water
- slide over the handles over the adapter tubes (2 sizes)

- slide them over the steer handles in place
- align the left mirror with the protruding nob point outwards and parallel with the steering, mark and cut a groove to accommodate indicator lights (use grease)
- fasten them in place while keeping in mind it also fastens the rubber handles

Step 16; installation of mudguards (0,5h)

Materials;

- 24x M5X16
- 24X M5 washers (5,3x15)
- 24x M5 nylon stop nuts
- place the individual reshaped mudguards over the axle and brakes.
- attach with the screws and tighten accordingly
- leave the front-left mudguard unattached for installing the parking brake.